Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 47 Friction Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 47 Friction Numerical

1 CLASS 47 Friction Numerical

Study focus: CLASS 47 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

The first question for today's session, question number six slash eight.

Key details

- determine the coefficient of kinetic friction that is which will allow the homogeneous body to move down the thirty degree incline at constant speed.
- There is an inclined plane.
- Inclination of the plane with the horizontal is thirty degree.
- And this is a homogeneous body and we have to assume here it is.
- A two dimensional body.
- One end of this body is resting on a roller and the other end is resting directly on this inclined plane.
- Since this homogeneous body is resting on an inclined plane, here we are going to assume X axis along the inclined plane.
- We know y axis will be normal to x axis.
- Next, since we are considering this homogeneous body.
- It will be having a weight and we know weight will be always vertically downward.
- This W is an inclined force with respect to XY plane.
- Either with respect to x axis or with respect to y axis.
- Since this is thirty degree here, the angle between w and y axis will also be thirty degree.
- will be W sin 30 degree and the y component will be downward and magnitude will be W cos 30 degree.
- As I said before, this is a roller.
- Since the roller is.
- On an inclined plane we are going to neglect the frictional force.

- Normal to x axis and I'm going to call it as NA.
- That is the normal reaction at A.
- Next comes this part.
- It's like a foot here which is in contact with the incline.
- It will be having both frictional force as well as the normal reaction.
- I'm going to call it as N B.
- That is normal reaction at B.
- And because of this incline this.
- Homogeneous body will have a tendency to slide down the plane.
- To avoid that the frictional force will be up the incline.
- This is the frictional force at B.
- We can use the formula mu k is equals to F B by N B.
- So we need to find first what is frictional force at P and what is normal reaction at B.
- To find these two we have dot conditions of equilibrium.
- So first one is this f b is towards the positive side of x axis, so plus f b.
- This W sine thirty since it is in the opposite direction, it is minus W sine thirty.
- These are the only two forces which are along the X direction.
- C we will get the value of F B.

Example or application from the lecture

- And the figure is given like this.
- So first thing is we have to consider here X and Y axis.
- Next we need to consider y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

Point five times that we can write it as mg and we know sign that is point five so FP will be equals to point five times set of mg.

Key details

- Call it as equation one.
- Since we need to find mu k we need to.
- Know the value of F B and N B.
- It's not required to know the value of N A.
- Sum of the moments of all the forces about A as 0.
- That is the next condition of equilibrium.
- Moment due to F B will also be zero because it is a.
- the point is in the same line of action.
- And N B is acting in this direction.
- It is counterclockwise.
- Moment due to this normal reaction at B.
- That's what is written here.

- Next comes this W cos thirty.
- It is acting in the clockwise direction, so it should be taken as minus.
- Minus W cos thirty into perpendicular distance will be due to symmetry.
- This distance will be B by 2.
- Next comes this force.
- It is also clockwise.
- And the distance.
- We are talking about this distance.
- Again, due to symmetry, this distance will be B by 2.
- That should be equal to 0.
- And the equation is going to be N B will be equals to.
- Point six eight three times the dup M G and call it as equation two.
- We know next mu case.
- Yes, B by N B.
- This is the coefficient of kinetic friction between the homogeneous body and this incline for which this body will be under motion with the uniform speed.
- Question number six slash thirteen.
- The fifty kg wheels on its hub up the circular incline under the action of the twelve kg cylinder attached to a card around the rim.
- Determine the angle at which wheel comes to rest, assuming that friction is sufficient to prevent slippage.
- What is the minimum coefficient of static friction which will prevent this position to be reached with no slipping?
- here you can look at this figure.
- This is the fifty kg field, which is attached to a twelve kg cylinder.
- And due to this twelve kg cylinder, this fifty kg wheel rolls on this circular incline.
- Because here the friction between this circular wheel and this incline is sufficient for this to be in a state of rest.
- At the same time we need to find the coefficient of friction here between this 50 kg wheel and this circular incline.
- That's what we need to find.
- Since the weight of this circular wheel is 50 kg.

Example or application from the lecture

- Ah, which is the next condition of equilibrium we should consider.
- No need to consider sigma f y is equals to zero because we have two unknowns along y direction, one is n a and one more is n b.
- Distance here it is given as B.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

Oh sorry, wheel is fifty KC that will be acting vertically downward.

Key details

- Fifteen to nine point eight one that is the weight of this wheel vertically downward.
- Next this wheel is in contact with here, so there will be a friction as well as the normal reaction.
- Since this will have a tendency to move this direction, the frictional force will be in the opposite direction, F will be up the incline.
- So next the same thing is considered here.
- there is a 12 kg cylinder attached to this wheel and the weight since it is freely suspended here the weight is going to be downward vertically downward to one into 9.81 N.
- I'll call this point as O and I'm going to apply the first condition as sum of the moment of all the forces about O as zero.
- This I'm going to start from here.
- It is acting in the clockwise direction with respect to this point of contact.
- So clockwise means negative, force is twelve into nine point eight one into distance.
- What is the distance between horizontal distance between this point of contact and the line of action of this force.
- That should be the horizontal distance, this distance.
- How to find this distance, if you look at this figure, the distance from the center of this wheel.
- Till here it is a radius that is two hundred millimeter.
- I am going to get that horizontal distance between the force and the moment center.
- This is the moment due to the 12 kg cylinder.
- Next moment due to F.
- And n will be zero because they are acting at the same point.
- self weight of the wheel since it is acting in the counterclockwise direction it should be taken as plus.
- This is W from here we know this is W into distance will be horizontal distance because the line of action is vertical the distance should be horizontal.
- I am going to call that distances.
- That should be equals to zero.
- Once I know the value of X1 here, then we will get the value of theta.
- Ah so how to find this X1?
- Is it possible to express X1 in the form of theta?
- Yes we can do that.
- Same AB V is taken here.
- If this angle is theta, then this angle will also be.
- Next A B as I said before that is X1.
- Then A U distance.
- A u is nothing but the radius of the smaller circle.
- The radius of the smaller circle we know is 75.
- sine theta, sine theta is equals to opposite by hypotenuse.
- X1 will be 75 sine theta.
- Substitute the value of X1 in the equation.

- Once we substitute the value of x1 in the equation one, we will get it as sin theta is equal to 0.516.
- Once we simplify this we are going to get the value of theta.
- Theta is thirty one point zero seven.
- Nothing but when the value of theta is thirty one point zero seven, the wheel which is rolling up this incline will be nest of.
- Rest because at this angle the frictional force here will be sufficient to prevent this from slipping under the motion.
- Next we need to find the value of F, coefficient of friction.
- For that we need to know the value of F and N.
- Once we know F.
- And n then mu will be is equal to f by n.

Example or application from the lecture

- Let's start with the first condition of equilibrium.
- I'm going to consider moment about this contacting point as zero.
- If you consider that the first one is.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 15:00–19:59

Sigma f of x is equals to 0.

Key details

- Since this f is along this x axis itself, it should be taken as plus.
- F, so plus F, then W will be inclined force with respect to X and Y.
- And the inclination here will be since this angle is theta, angle with respect to Y axis will be theta since we are talking about x component of that's going to be w sine theta and the component is going to be left side so minus w sine theta.
- That should be equal to zero.
- Already we know the value of theta from the previous.
- Next, we should know the value of normal reaction.
- So we need to apply the second condition of equilibrium, sigma f of y is equal to 0.
- The y component of this W will be downward like this.
- Minus W into cos theta will be the component.
- Then Y component of this will be again downward.
- We know the value of theta, we need to find only the value of n.
- Once we know.
- Frictional force in the normal reaction, then coefficient of friction will be equal to F.
- We will get it as point six zero two.
- This is the answer.

Example or application from the lecture

- Here we are going to consider incline as like x axis will be along this incline like a tangent here.
- Then minus of we need to consider this, the x component of this will be again left side.
- When we consider sigma f y 0, n is upward, so plus n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 47 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice